Content Analysis in Consumer Research

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The use of documentary evidence such as historical records, novels, existing advertisements, and photographs has been little used in consumer research. This paper presents an exposition of content analysis methodology and, tangentially, reviews the existing studies in consumer behavior using this rich approach.

In 1953, more than two decades ago, Cartwright wrote:

> When one stops to think of it, it is really surprising how much of the subject matter of social psychology is in the form of verbal behavior. The formation and transmission of group standards, values, attitudes, and skills are accomplished largely by means of verbal communication. Education in the schools, in the home, in business, in the neighborhood, and through the mass media is brought about by the transmission of information and by the exercise of controls which are largely mediated through written or spoken words. If one is concerned with problems of social organization, the situation is similar. Supervision, management, coordination, and the exertion of influence are principally matters of verbal interaction. Social and political conflicts, although often stemming from divergent economic interests and power, cannot be fully understood without studying the words employed in the interaction of conflicting groups, and the process of mediation consists largely of talking things out. The work of the world, and its entertainment too, is in no small measure mediated by verbal and other symbolic behavior. (Cartwright 1953, p. 422)

Surprising or not, there is no doubt that much of the subject matter of the social sciences including consumer studies is in the form of verbal and symbolic behavior. The exchange process in the marketplace and the communication of the values of the exchange depends upon the written or spoken word. And yet, research on communications content, the "systematic description of these phenomena," has not been accorded much attention in consumer research although first introduced to the field two decades ago by Ferber and Wales (1958).

Although not independent, the study of content variables is approached apart from the study of the communicator or the audience. The signs and symbols (Mead 1934; Morris 1946) are the units of analysis rather than the intent of the communicator or the actions of the interpreter. Of interest is what was said, the properties of the stimuli, rather than what the communicator claims he said or the interpreter perceived to have been said. Much of consumer research has concentrated on the characteristics, opinions, or behavior of the interpreter of communications messages or on the characteristics of the communicator. Content analysis is the study of the message itself, and not the communicator or the audience. It is the study of the stimulus field (Fearing 1953). The purpose of this paper is to present and discuss some of the issues in content analysis methodology and, tangentially, to review the studies carried out in consumer research using the technique.

DEFINITIONS OF CONTENT ANALYSIS

Content analysis, although relatively new to consumer research as a formal methodology, is not new to political science, journalism, social psychology, communications research, or political propaganda analysis. The literature in these fields is replete with detailed studies and definitions. For example:

- Content analysis is a research technique for the objective, systematic, and quantitative description of the manifest content of communication. (Berelson 1952, p. 55)
- Latent content as well as manifest content may be examined by content analysis, a series of judgments or descriptions made under specifically defined conditions by judges trained in the use of objectively defined criteria. (Fearing 1954)
- The term "content analysis" is used here to mean the scientific analysis of communications messages. . . . The method is, broadly speaking, the "scientific method," and while being catholic in nature, it requires that the analysis be rigorous and systematic. (Barcus 1959, as cited in Holsti 1969, p. 3)
- Content analysis is a phase of information-processing in which communications content is transformed,

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through objective and systematic application of categorization rules, into data that can be summarized and compared. (Paisley 1969, p. 133)

- Content analysis is a systematic technique for analyzing message content and message handling—it is a tool for observing and analyzing the overt communication behavior of selected communicators. (Budd, Thorpe, and Donohew 1967, p. 2)
- Content analysis, while certainly a method of analysis, is more than that. It is . . . a method of observation. Instead of observing people's behavior directly, or asking them to respond to scales, or interviewing them, the investigator takes the communications that people have produced and asks questions of the communications. (Kerlinger 1964, p. 544)
- Content analysis will not tell us whether a given work is good literature; it will tell us whether the style is varied. It will not tell us whether a paper is subversive; it will tell us if the contents change with party line. It will not tell us how to convince the Russians; it will tell us what are the most frequent themes of Soviet propaganda. (Lasswell, Lerner, and Pool 1952, p. 45)

These researchers and others agree that the distinguishing characteristics of content analysis are that it must be *objective*, *systematic*, and *quantitative*.

Objectivity

The requirement of objectivity stipulates that the categories of analysis be defined so precisely that different analysts may apply them to the same body of content and secure the same results (Berelson 1952).

Each step in the research process must be carried out on the basis of explicitly formulated rules and procedures. Even the simplest and most mechanical forms of content analysis require the investigator to use his judgment in making decisions about his data. What categories should be used? What criteria are to be used to decide that a content unit (word, theme, story, picture, etc.) should be placed in one category rather than another? From these data what was the reasoning that led to one inference rather than alternative ones? According to Holsti (1968), objectivity implies that all decisions are guided by an explicit set of rules that minimize—although probably never quite eliminate—the possibility that the findings reflect the analyst's subjective predispositions rather than the content of the documents under analysis. Thus one test of objectivity is: Can other analysts, following identical procedures with the same set of data, arrive at similar conclusions?

This requirement of objectivity gives scientific standing to content analysis and differentiates it from literary criticism. A newspaper reporter studying Soviet press coverage of an event, a politician carefully listening to the speeches of his opponent, or a TV critic discussing coverage on American television of the same event cannot be classified as employing scientific method. Their results may be penetrating, sensitive, shallow, accurate, literary, or biased, but not scientific. For what is lacking is replicability, reliability, or what Berelson has chosen to call objectivity.

In the great majority of studies in the consumer behavior literature where some form of content analysis has been carried out, the objectivity requirement has seldom been met. Typically, the author has analyzed the communications material himself with no expressed concern about the reliability of the analysis or controls for selective perceptions and biased predispositions.

Systematization

Systematization means that the inclusion and exclusion of communications content or analysis categories is done according to consistently applied rules (Holsti 1969). This requirement is meant to eliminate partial or biased analysis in which only those elements in the content which fit the analyst's thesis are selected. The requirement clearly eliminates analysis in which only materials supporting the investigator's hypotheses are admitted as evidence.

The second meaning of "system" is that analysis must be designed to secure data relevant to a scientific problem or hypothesis (Berelson 1952). The findings must have theoretical relevance and be generalizable. Purely descriptive information about content unrelated to other attributes is of little value. Thus, a tabulation simply reporting the number of periodicals acquired by a librarian would not represent a content analysis study unless the results were used for a trend or comparative analysis or for some other generalization. Stated somewhat differently:

> a datum about communications content is meaningless until it is related to at least one other datum. The link between these is represented by some form of theory. Thus all content analysis is concerned with comparison, the type of comparison being dictated by the investigator's theory. (Holsti 1969, p. 5)

Quantification

This requirement is perhaps the most distinctive feature of content analysis. Quantification of judgments distinguishes content analysis from ordinary critical reading. A measurement of the extent of emphasis or omission of any given analytic category is what content analysis is all about. Although early researchers equated quantification with strict frequency count (Leites and Pool 1942, as cited in Lasswell, Leites, and Associates 1949; Janis 1943) and the assignment of numerical values (Kaplan and Goldsen 1949), this demand is perhaps too strict and unnecessary (Kracauer 1952). Berelson (1952) feels that the quantification requirement can take the form of quantitative

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words like more, always, increases, or often. What is implied by the quantification requirement is that the data be amenable to statistical methods not only for precise and parsimonious summary of findings but also for interpretation and inference. Whether the statistical method involved requires parametric or nonparametric data or nominal, ordinal, or interval scales is not relevant.

THE USE OF CONTENT ANALYSIS

To summarize, content analysis is a scientific, objective, systematic, quantitative, and generalizable description of communications content. In consumer behavior and marketing, content analysis has been used to study the following questions:

- What are the changing values in society as reflected in the analysis of mass periodical fiction (Johns-Heine and Gerth 1949)? It the Riesman hypothesis about increasing other-directedness of American society supported by changing content of consumer goods advertising (Dornbush and Hickman 1959)?
- What are the product and company images of selected consumer goods as reflected in the mass media (Stone, Dunphy and Bernstein 1966; Woodside 1972)?
- What are the advertising appeals that are used for nontechnical graduates in a college newspaper (Ybarra 1970)? Do recruiting appeals for technical and managerial positions in newspaper advertisements differ in their inner- and other-directedness (Makinson and Welge 1970)?
- What are the content characteristics of best-selling novels and can literary success be predicted by analyzing content variables (Harvey 1953)?
- Which of several decision-choice models (compensatory, lexicographic, risk, etc.) are used by magazine and television advertisers (Wright and Barbour 1975)?
- Are minority Americans presented in a prejudicial fashion in magazine fiction (Berelson and Salter, 1946)? Have minorities on television gone through successive stages from nonrecognition to ridicule to respect as suggested by social science theory (DeFleur 1964)?
- Does the portrayal of blacks differ between television, magazine, and newspaper presentations (Cox 1970)?
- What is the frequency of appearance and roles of blacks and other minorities in the mass media (Shuey, King, and Griffith 1953; Boyenton, 1965; Cox 1969; Kassarjian 1969, 1976; Dominick and Greenberg 1970; Ferguson 1970; Roberts 1970; Wanderer 1970; Geizer 1971; Greenberg and Kahn 1971; Bush et al. 1974, 1977; Hair et al., 1977)?
- What is the portrayed image and role of women in the mass media (Courtney and Lockertz 1971; Wagner and Banos 1973; Venkatesan and Losco 1975; Belakaoui and Belakaoui 1976)?
- How are women, blacks, foreigners, and conservatives depicted in comic strips (Spiegelman, Terwilliger, and Fearing 1952, 1953)?
- What is the information content found in television advertising (Resnik and Stern 1977)? Is comparison advertising leading to misleading, dysfunctional, and ambigious messages on television (Shimp 1975)?
- What is the ease of readability of various marketing, advertising, and consumer research journals, including *The Journal of Consumer Research* (Lacho, Stearns, and Villere 1975)?[1]

Naturally, a wide range of other problems can be studied by content analysis ranging from measures of bias in newscasting and news reporting to differences between communications in the stylistic use of verbs, nouns, adjectives, and adverbs. According to Holsti (1969), content analysis is likely to be especially appropriate for at least three general classes of research problems which may occur in virtually all disciplines and areas of inquiry.

First, it may prove useful when data accessibility is a problem and the researcher's data are limited to documentary evidence (e.g., the social and organizational structure of Soviet prisons, McDonough 1975) or under conditions when subjects can no longer be easily located (e.g., "Public Opinion in Colonial America," Merritt 1963). Moreover, content analysis may also function as a supplementary source of data or as an unobtrusive measure (Webb et al. 1966; Webb and Roberts 1969). For example, in a study of public attitudes and voting intentions, researchers at Stanford University have used not only survey data involving interviews and questionnaires, but also carried out content analyses of "letters to the editor" (Roberts, Sikorski, and Paisley 1967), political bumper stickers, graffitti, and defaced political posters.

Such multiple indicants for hypothesis testing add a dimension of validity seldom found in consumer behavior research. In other cases, content analysis as an unobtrusive measure may be necessary when direct ob-

[1] Using the Flesch index based on average sentence length and number of syllables per 100 words, the *Journal of Consumer Research* in 1974 ranked the most difficult to read among the 11 journals studied. The readability scores were lower (harder to read) than the *Journal of Advertising Research*, *Journal of Marketing*, and *Journal of Marketing Research*, among others.

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servation might alter the behavior pattern of the subjects, such as in "love" and "changing courtship patterns" (Carey 1968; Horton 1956; Webb and Roberts 1969).

Second, some form of content analysis is necessary when the subject's own language and mode of expression is crucial to the investigation. Analysis of information processing protocols, responses to projective tests, or subtle inferences to be found in in-depth interviews or focus groups are examples.

Finally, content analysis can be particularly helpful when objective systematic evaluation of material is needed that is much too voluminous for a single investigator. In studies of newspapers, movies, radio, literature, Soviet propaganda, Voice of America themes, or the role of women in magazine literature, the volume of material may well exceed the investigators ability to undertake the research himself, no matter how unbiased or aware of his own selective perception he may be. Under conditions in which the training of assistants, systematic sampling of stimuli, and interjudge reliability become critical, the formal methods of scientific content analysis become indispensable.

Nevertheless, content analysis is not relevant to all research. It can rarely be used to determine the truth of an advertising claim, or to evaluate the aesthetic qualities of a TV program. Content analysis cannot determine whether or not Ban antiperspirant in fact "keeps you 20% drier," or whether Farrah Fawcett-Majors will be perceived as "incredibly beautiful." Also, it might be foolish to use content analysis to determine whether the *New York Times* supported Humphrey or Nixon in 1968 when methods other than content analysis could be used more efficiently (Holsti 1969). However, if we wished to determine at this late date whether the *Berliner Zeitung* supported the rise of Hitlerism in the 1920's or whether the rise of Hitler could have been predicted from the themes found in German motion pictures (Kracauer 1947), content analysis might well be necessary.

The procedures involved in the methodology consist of selecting, from the available population of documents to be studied, a reasonably sized sample for study. The second step would be to determine the unit of measurement, whether it be the specific word, an overall theme, or simply the existence or nonexistence of some event or claim. Then the procedures call for the training of judges for categorization of the content according to predetermined rules, and finally, statistical treatment and analysis of data.

SAMPLING

For most content analysis studies, the immense task of analyzing existing documents begins with sampling procedures. Suppose one were to study the role of women in network television prime-time advertising during 1977. One can quickly imagine the enormous number of ads that would have to be analyzed. Further, if the research question were something like, "A comparison of local and network television advertising presentations of the role of women in society," the number of ads to be processed would have to be increased many fold. If, in addition, one were also interested in differences over time, regional or geographic differences, or comparisons between prime-time and daytime television, the need for sampling and careful definition of the universe, rather than analysis of the entire population, becomes painfully evident.

Fortunately, researchers in the fields of marketing and communications as well as in consumer research are acquainted with sampling procedures. Most of us have been trained in drawing samples from populations of consumers, or people in general. Although drawing a sample of documents may pose some unique problems, the concerns of the research are no different. Needed is a sample of manageable size, randomly drawn, that is representative of the defined universe such that generalizations are possible. The sample rather than consisting of consumers, voters, or subscribers, is composed of comic strips, magazine advertisements, daily newspapers, or scripts of radio commercials.

Other than simple random or interval sampling, sophisticated procedures such as cluster sampling and multistage sampling may be appropriate. The problems in determining what size sample to use are similar to determining sample size in other aspects of consumer research.

UNITS OF MEASUREMENT

Content analysis calls for the qualification of elements in the communication stimuli. An element or subdivision of the content may range from large to small. Thus a content analysis could determine the extent of support or opposition on a controversial issue found in local newspapers. The analysis could examine the amount of space devoted to the topic or the number of articles, a sampling of the paragraphs or sentences in the articles, or even selected key words or terms. Such subdivisions are the units of analysis.[2]

Word

The smallest unit generally used is the word. According to Berelson, content analysis can be done on the basis of a single letter as the unit in cryptoanalysis and the breaking of international codes. Such activity, however, is usually beyond the scope of consumer research. The word as a unit is identical with what Lasswell (1952) calls a symbol and may include

[2] Perhaps the finest presentation of this material is to be found in Berelson (1952). Most authors in the field of content analysis have based their presentations on Berelson in discussion of the units of content analysis. This paper continues in that tradition.

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word compounds, e.g., phrases, as well as single words. In this type of research one might study the relative occurrence of key symbols or value-laden terms such as religious, sexual, supernatural, democratic, social, friendly, clean, sparkling, and so on, until the content has been systematically examined relevant to the hypotheses of the study. Readability studies (Flesch 1951) have used the word unit in content analysis. Typically, such studies construct readability formulas on the basis of personal references, prepositional phrases, different hard words, words with prefixes and suffixes, etc. (Berelson 1952).

Theme

The next larger unit is the theme, a single assertion about a subject. The theme is among the most useful units of content analysis because issues, values, beliefs, and attitudes are usually discussed in this form. However, it is also the most difficult unit of analysis. For example, the sentence, "These clandestine Soviet actions on the imprisoned island of Cuba will not be tolerated by the American people," contains assertions about three nations. The coder must be able to reduce this sentence into its component themes before they may be placed in the proper categories (Holsti 1968).

Perhaps the best known thematic analysis in consumer behavior was made of the content of Kate Smith's broadcast on the marathon bond drive during World War II. After 18 hours of hourly and half-hourly spot announcements from fifteen seconds to one minute each in length, Kate Smith managed to sell 39 million dollars worth of War Bonds through the CBS radio network. The content of her appeals was analyzed by themes—those of sacrifice, participation, competition, facilitation, and familial and personal appeals appearing most often (Merton 1946).

Character

Use of the fictional or historical character as the recording unit is often employed in the studies of fiction, drama, movies, radio, and other forms of entertainment material. In a well-known study involving the appeals of the radio daytime serial or soap opera, Arnheim (1944) used the characters as the unit of analysis. A fascinating study on ethnic characteristics of heroes and "bad guys" in Sunday comics, using the character as the unit of analysis, can be found in Spiegelman, Terwilliger, and Fearing (1953).

Item

The item is the whole natural unit employed by producers of symbolic material. It may be the entire speech, radio program, letter to the editor, editorial, or news story. This unit is often too gross for most research according to Holsti (1969), but Berelson (1952) presents two dozen studies in which the unit of analysis was the item—motion pictures, musical compositions, paintings, cartoons, and jokes. Most consumer behavior studies previously mentioned as using content analysis have used the item as the unit of analysis. For example, the studies on the role of blacks or the role of women in American advertising have used the entire advertisement for various aspects of the analysis.

Space-and-Time Measures

Some studies have classified content by physical divisions, such as the column inch (newspapers), the line or paragraph, the minute (radio and TV), or the foot (films). Almost every conceivable space-time measure has been used.

In a given study, different units will sometimes be employed to test each of several hypotheses. Thus the item (total ad) may be the unit of analysis to examine the frequency of the use of blacks in magazine advertising. A space-time measure may be employed to determine the relative size of "black" ads as compared to "white" ads. The theme may be utilized in analyzing the values presented by the communicator, and a word count used as a measure of emphasis.

CATEGORIES OF ANALYSIS

Content analysis calls for the categorization of the various elements in the occupations of the characters in the novel, the frequency of occurrence of incompetent females in detergent advertisements, or the proportion of racially integrated advertisements in general circulation magazines.

Content analysis is no better than its categories, since they reflect the formulated thinking, the hypotheses, and the purpose of the study. The categories are, in essence, the conceptual scheme of the research design. For example, Wayne (1956) attempted to compare the values expressed in the content of two major family magazines, one in the United States (*Life Magazine*) and the other in Soviet Russia (*Ogonek*). His purpose was to shed light on the different underlying value systems of these two powerful and contending nations.

Life and *Ogonek* were the leading general circulation weekly pictorial magazines that appealed to most members of the family. The universe was restricted to pictures and pertinent captions, if necessary, rather than the text. Hence, the purpose was a study of American and Soviet themes and values, as depicted in pictures presented in thirteen randomly selected issues of both magazines in the year 1948. A further limitation was imposed on the sample by excluding from the analysis all pictures lacking human action.

The categories selected for the study of values were an adapted version of Spranger's value categories from his *Types of Men* (1928). The definition of the categories emerged from Spranger's treatise. For example, in

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order to come under the *theoretical* category, a picture had to show an activity connected with the search of truth. Thus a photo showing scientists at work might be so classified. A photograph of engineers and technicians laboring in their laboratories to increase harvests in the Ukraine or the American prairie might be categorized as *economic* (utilitarian) value. The results indicated a greater incidence of economic and aesthetic values in the Soviet magazine and greater emphasis on religious and social values in the United States. There were no differences between the two countries on theoretical or political (power) values.

In this particular study, the judges were to classify each picture into one of several polynary categories (e.g., economic, theoretical, religious, social, political, and aesthetic). Other researchers have used a binary method of classification. In one phase of their study, Spiegelman, Terwilliger, and Fearing (1952) wished to examine the situations in which comic strip adventures occurred. The authors arranged the categories in a series of dichotomous decisions to be made by each judge. Is the action on Earth or in interstellar space? If on Earth, is it in the United States or is it foreign? If in the United States, is it historical or contemporary? If contemporary, is it rural or urban?

Whether the categories should be binary or polynary has been studied (Schutz 1952, 1958) with the conclusion that the polynary approach generally results in better reliability figures but only for the more difficult decisions.

Consumer researchers, experienced in questionnaire construction and in developing coding categories for open-ended questions, will find similar problems and procedures in creating content analysis categories. In the former case the respondent presents his opinion. The function of the interviewer or the coder-editor is to convert the verbal statement into a classificatory scheme. In the latter case, a highly trained judge, following explicitly stated rules, selects the categories that best describe the content.

Direction

The problem of direction is probably one of the most frustrating issues facing the researcher because it is one area in which the element of subjectivity is difficult to control and impossible to eliminate entirely (Budd, Thorp, and Donohew 1967). Direction refers to the pro or con treatment of the subject matter. Basically, it asks the question: Is the communication for or against the particular subject, or neutral? In the literature various synonyms have been used for pro and con, presumably reflecting differences in the specific materials analyzed: indulgence-deprivation, approval-disapproval, favorable-unfavorable, positive-negative, for-against, and optimistic-pessimistic. Obviously, content analysis can be most productive when it is able to show direction—or the lack of it. "Although direction is a commonly recognized characteristic of communications content, it is not always easily analyzed in an objective fashion. Many textual passages are not clearly pro or con or neutral; the borderline is often indistinct" (Berelson 1952, p. 150).[3]

To exemplify the issues and problems in selecting the units of measurement and the categories of analysis one can consider the following paragraph:

> I'd like to see all trade barriers down after the war. Raw goods should be shared where they're needed. It's money and raw goods and poor living that caused most of this war. We should see that Germany gets its fair share this time or we'll have another war. Russia is fighting for her way of life just like we are for ours. England is fighting along with us and Russia to protect the people against fascism—to be free, not slaves. Churchill and Roosevelt and Stalin are great men. They know how the people feel. We can't stay on our side of the pond anymore. The union's taught me that. (Cartwright 1953, p. 435)

Literally dozens of variables or attributes can be found in this passage. To list a few: number of words, percentage of personal pronouns, attitude toward free trade, perceived cause of war, degree of confidence in the Allies, degree of confidence in leaders, attractive traits of leaders, attitude toward isolationism, evidence of previous isolationism, source of influence on attitudes, implied values, inclusiveness of cognitive structures, and degree of approval of war aims. Once the variables under study have been determined, say confidence in leaders, the variable under scrutiny must be broken down into categories, such as unqualified confidence in leaders, qualified confidence, balanced, qualified mistrust, unqualified mistrust, not classifiable (Cartwright 1953). Obviously, not only are many other attributes or variables possible depending on the hypotheses of the study, but many other schemes for categorization are also possible.

THE PROBLEM OF RELIABILITY

Since the researcher's subjectivity must be minimized to obtain a systematic, objective description of the communications content, the issue of reliability becomes paramount. Reliability, or reproducibility, is one of the distinguishing characteristics of content analysis as contrasted with other techniques of describing communications content—the work of a literary critic, the commentary of a newspaper reporter, or the opinions of a layman. "The importance of reliability rests on the assurance it provides that the data obtained are independent of the measuring event, instrument, or person" (Kaplan and Goldsen 1949).

[3] The rather elaborate rules that have been devised for classifying direction are beyond the scope of this paper. The interested reader should turn to Budd, Thorp, and Donohew (1967), Janis and Fadner (1949), and references therein.

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Category Reliability

This important measure of reliability depends upon the analyst's ability to formulate categories and present to competent judges definitions of the categories so they will agree on which items of a certain population belong in a category and which do not. If this aim cannot be achieved (and the other sources of disagreement are controlled), the understanding of the category as specified is not yet sufficiently clear for scientific usage (Schultz 1958, p. 512).

In the Kassarjian study on the role of blacks in magazine advertising (1969, 1976), one of the categories consisted of racial composition of the characters in the advertisement. The categories consisted of *all black* characters, *all caucasian* characters, *integrated peer* (in which both black and white characters are peers), *integrated nonpeer, separate pictures* (panel ad in which the characters do not appear in the same scene), and *don't know*. The reliability of this category was nearly 100%, that is, judges did not disagree about whether ads contained all black models, all white, or both in a peer or nonpeer relationship.

Perusal of ads that contained both black and white characters in a nonpeer relationship indicated that one seldom saw a white shoeshine boy with a black patron or a black supervisor giving orders to a white employee. A category was then created called the authority relationship. The author was interested in finding the frequency of ads in which the black model had a superior role to the white. Although the meaning of this category was clearly described, defined, and redefined, and the judges trained and retrained, reasonable levels of reliability could not be established.

In many ads the judges could not agree on which character had the superior role. For example, one ad contained white patrons in an exclusive restaurant sitting at a table with expensive linen and silver service. A black waiter with a haughty patronizing glare was waiting for orders. Some judges claimed the white patrons were in the superior role, giving orders to the black waiter. Other judges claimed the intimidating waiter was clearly in the superior role and that the patrons were pretty well at his mercy. After fruitless effort to reach agreement, the category was discarded with the explanation that superior and inferior roles in interpersonal relationships between white and black could not be scientifically defined in that study.

Interjudge Reliability

Interjudge reliability is the percentage of agreement between several judges processing the same communications material. It is the degree of consistency between coders applying the same set of categories to the same content.

A commonly used measure of reliability is the ratio of coding agreements to the total number of coding decisions. Thus, if in a particular study two judges make a total of 1,000 decisions each, and agree on 930 of them and disagree on 70, the coefficient of reliability would be 93 percent. If more than two judges are involved, the typically reported figure is the percent agreement between each pair of judges. The average interjudge agreement can be presented as a composite reliability score. A second approach to a composite score would be to take a ratio of all coding agreements to the total number of coding decisions made by all judges.[4]

Generally, the reported reliabilities in the literature are extremely high. Berelson (1952) claims the range is between 66 and 95 percent with a concentration at about 90 percent. It is my belief that researchers can be quite satisfied with coefficients of reliability above 85 percent. Studies with reported reliabilities of less than 80 percent should be treated with suspicion.

Fortunately, there is a reasonable likelihood of underestimation of reliability in many studies, arising from the measurement of reliability on detailed categories for reporting purposes (Berelson 1952). For example, in the Kassarjian study of the role of blacks in advertisements, the greatest disagreement emerged from two or three variables. One of these was the judgment whether the advertisement was a photograph of an actual scene, a drawing, or cartoon. Judges had difficulty discriminating between a drawing and a cartoon. Nevertheless, the coefficient of reliability was 91.8 percent. In reporting the data these two categories were combined, as were others that caused difficulty in discrimination. As in most studies reported in the literature, a new or net reliability figure based on the combined categories was not tabulated.

As can be seen in this example, if the categories are narrowed and simplified, and the decisions for the judges made quite simplistic, reliability can be increased, but at a price. "Reliability is a necessary condition for valid inquiry, but paradoxically, the cost of some steps taken to increase reliability may be a reduction in validity" (Holsti 1969). In formulating research designs, analysts have often been forced to strike some balance between reliability and the relevance of categories and units. Obviously, the reliability coefficient cannot be the sole criterion for the quality of a study.

[4] Using for a reliability index the percentage of judgments on which coders agree out of the total number of judgments, has been criticized by Scott (1955). By chance alone, agreement should increase as the number of categories decreases. One would expect better agreement on a two-category variable or scale than on a five-category variable. To correct for this, Bennett, Alpert, and Goldstein (1954) and Scott (1955) have presented new reliability indices. However, the literature continues to follow the more conventional approaches to reporting reliability figures, and hence technical details of the Scott approach will not be discussed in this paper.

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VALIDITY

Validity is defined as the extent to which an instrument measures what it purports to measure. Any time an inference is made from the results of an instrument, questionnaire, scale, or other tool, questions of reliability and validity should be paramount. In the field of content analysis, choice of categories and content units enhances or diminishes the likelihood of valid inferences.

The methodological literature abounds in references to validation procedures, ranging from content validity to predictive, concurrent, and construct validity measurements (Technical Recommendations 1954) to the multiple indicant approaches of Webb, et al. (1966), Webb and Roberts (1969), and Roberts, Sikorski, and Paisley (1969).

Unfortunately, the few content analysis studies that exist in the consumer research literature have completely ignored measures of validity. Perhaps the best that can be expected at this stage of development of content analysis in consumer research is that after close examination of the units of measurement, categories, sampling procedures, and results, the instruments appear valid using some form of content validation or face validity. Over time one could expect that with repetition of studies and multiple measures of similar hypotheses from several points of view with multiple tools, differing coding schemes, and varied sources of communications content, reasonable measures of validity may emerge in the consumer behavior literature. In the development of categories and other features of the methodology, researchers in the future might build into the study design measures of predictive, concurrent, and construct validity.

THE COMPUTER

Often content analysis involves an enormity of dull clerical processing of data, and yet requires highly trained sensitive coders. Doubtless, reliability, accuracy, validity, and the ability to make inferences and generalize from the data suffer if bored coders do not carry out their tasks accurately.

In this day and age, when tasks require repetitive and tedious behavior, the natural solution is to turn to machines. Also in content analysis numerous attempts have been made to turn to the computer. The involvement has gone beyond simple tabulations of frequencies and data manipulation to having the computer do the actual coding. The machine can be used in any task for which unambiguous instructions can be prepared.

The greatest use of computers has been in those studies that require a word count or symbol count. Computers can count words at high speeds and with perfect reliability. Numerous programs exist which convert written passages, novels, or entire languages into several hundred categories. The analyst's task consists of little more than keypunching the entire text on computer cards. The output serves as the basis for subsequent analysis and hypothesis testing.[5]

The General Inquirer and Dictionaries

Perhaps the most sophisticated use of computers in content analysis has emerged as part of the General Inquirer System. In the late 1950's, several investigators began working on the problems of automatic syntax analysis and synthesis, mechanical translation from one language to another, storage and retrieval of large amounts of text information, indexing, and the design of question and answer systems (Stone et al. 1966). From this work at MIT a set of computer procedures for processing "natural text" emerged; procedures that locate, count, and tabulate text characteristics.

The core of each General Inquirer System of content analysis is a dictionary in which each word is defined with one or more "tags" representing categories as ordinarily used in content analysis. The text to be analyzed is punched on computer cards, usually with some prior editing to separate complex sentence or thought sequences. The computer will then process the text into root words and tags or categories. Themes can also be processed, and programs have been prepared to deal with homographs (words with different meanings but identical spelling, e.g., "bear in mind that a male circus bear cannot bear a cub"). In short, the General Inquirer program can do much that human processors had to handle in the past.[6]

In recent years as advanced versions of the General Inquirer have emerged, the potential for computer-based research has become even more appealing. Studies in consumer research, however, have not yet been logically amenable to computer technology. In many cases published consumer research has tended to use television advertisements measured live off the airwaves or magazine advertisements in which the picture was critical to the study. Computers are not yet programmed to deal with nonverbal communications material such as a picture. In other instances, even where the unit of measurement is the sentence, theme, or word, the use of computers may not be practical. The available dictionaries, root words, and tags may not be appropriate, but of greater concern should be the

[5] Readers interested in the technical detail can turn to the several papers in Gerbner, et al. (1969), Stone, et al. (1966), and references to actual studies using computer output presented in Budd, Thorp, and Donohew (1967) and Holsti (1968, 1969). A paper by Starkweather and Decker (1964) may be particularly helpful.

[6] Technical details are available in Dunphy, et al. (1965); Hunt, et al. (1965); Stone, et al. (1962, 1965, 1966); and the several papers in Gerbner, et al. (1969).

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fact that preparation of the study for computer mechanization may well not be worth the effort involved. It is probably still easier to do it by hand, especially since bright graduate students with tolerance for clerical detail are not yet in short supply.

CONCLUSION

Bernard Berelson, one of the earliest researchers in content analysis, in another place and at another time, has written the following conclusion to one of his papers. It fits this paper, in this place, and at this time just as well:

> This, then, is content analysis. . . . How the contributions of content analysis measure up against those of other methods in social science research is difficult if not impossible to say. Because of the relative availability of the raw material, many sins have been committed in the name of content analysis simply because something else was harder or more inconvenient to do. At the same time, however, the method has produced a number of useful studies which combine original ideas with sound, careful documentation. This is, of course, the heart of the matter.
>
> In content analysis, as anywhere in social research, it is important to start in the right way. Simply going on a fishing expedition through some common communication material is almost certain to be unrewarding. Unless there is a sensible, or clever, or sound, or revealing, or unusual, or important notion underlying the analysis, it is not worth going through the rigor of the procedure, especially when it is so arduous and so costly of effort. (Berelson 1954, p. 518)

In consumer research, content analysis offers no new magical qualities. In fact, it demands that the researcher, rather than merely describing his impression of documents, be excruciatingly systematic, objective, and quantitative. Yet, the methodology does open new avenues for research, ranging from studies on the themes and appeals found in mass media to studies of deception in advertising.

The methodology allows for the reliable, valid, and quantitative answers to questions such as the following:

1. Have recent consumer oriented actions such as the affirmative disclosure program of the FTC been effective in changing the content of advertising? Has comparative advertising led to greater information being communicated to the consumer?
2. Do certain advertisers (or agencies) engage in the use of certain types of themes, appeals, claims, or deceptive practices more than other advertisers or agencies? What is their apparent intent? Who is their apparent audience?
3. Do themes and appeals found in television (or other media) differ from country to country? How do the press releases disseminated by the Exxon corporation in the United States, Israel, and the Arab countries differ? Do the themes, informational content, or questionable practices found in the advertising of multinational corporations differ from country to country?
4. Do certain newscasters (newsmagazines, newspapers) use more liberal (conservative) symbols and demonstrate more bias than other newscasters? Is editorial support for a political candidate also reflected in biased news sections?
5. In what ways do the rhetoric, articles, speeches, and books that emerged in the consumerism wave of the 1960's differ from the two earlier waves? Did the themes and appeals greatly change over the years?
6. What is the image of big business (government, universities) as reflected in advertising, television programs, radio disk jockey jokes, or magazine fiction?
7. What values are disseminated by comic strips such as Doonesbury, Peanuts, Mary Worth, Little Orphan Annie, Dick Tracy, Andy Capp, etc.? Has the portrayal of minorities, foreigners, liberals, women, children, or scientists changed in the past decades?
8. What values are reflected in the graffiti found in subway stations, buses, men's and women's toilets, various ghettos, or various countries? Can assumptions about the tensions, values, and concerns of a people be made by using unobtrusive measures such as analysis of graffiti, bumper stickers, jokes, or jargon?
9. What are the stylistic, content, and readability differences between media such as *True Confessions* and the *New Yorker*, between public service, local and network television, between consumerists and apologists for industry, or between textbooks, journals, or trade publications?

Further, using content analysis, one can make assumptions about readership of magazines no longer in existence, such as the *Saturday Evening Post*, or gather data about the apparent readership of an existing magazine without approaching subscribers. Analysis of historical documents could lead to conclusions about public opinion, consumer values, or buyer beliefs in an earlier era. Content analysis has been and can be used to prove authorship of particular documents, such as the Federalist Papers. The effect of public policy, as for example, advertising substantiation and new trade rules emerging from governmental agencies, can be tested for effectiveness. The extent of industry's "newly found" social responsibility can be examined.

In short, the methodology can be most useful whenever documentary evidence is available. Its limits are

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perhaps the limits of the ingenuity and creativity of the consumer researcher. That it has not yet been extensively used in studies of consumer behavior is really rather remarkable considering our proclivity to use varied methodologies and varied approaches to consumer issues.

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